

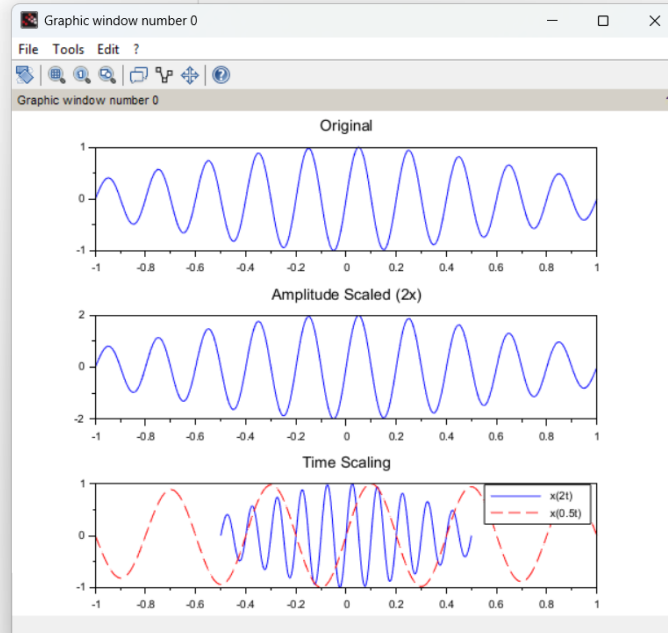
EXPERIMENT 2

ss 2.2.sci (C:\Users\naikn\Documents\ss 2.2.sci) - SciNotes

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```
1 t = -1:0.001:1;
2 x = sin(2*pi*5*t).*exp(-t.^2);
3 //Amplitude-scaling
4 x_amp = 2*x;
5 //Time-scaling
6 x_time_comp = interp1(t, x, 2*t, 'linear');
7 x_time_exp = interp1(t, x, 0.5*t, 'linear');
8 clf;
9 subplot(3,1,1);
10 plot(t, x); xtitle("Original");
11 subplot(3,1,2);
12 plot(t, x_amp); xtitle("Amplitude-Scaled: (2x)");
13 subplot(3,1,3);
14 plot(t, x_time_comp, 'b');
15 plot(t, x_time_exp, 'r--');
16 legend("x(2t)", "x(0.5t)");
17 xtitle("Time-Scaling");
18
19
```



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```
1 //Test-Case: Generation of Periodic and Aperiodic Signals
2 // Parameters for the periodic signal (Sinusoidal)
3 f_periodic = 1; // Frequency in Hz
4 A_periodic = 1; // Amplitude
5 t_periodic = 0:0.01:2; // Time vector for 0 to 2 seconds
6 // Generate the periodic signal
7 x_periodic = A_periodic * sin(2 * pi * f_periodic * t_periodic);
8 // Plot the periodic signal
9 clf; // Clear the current figure
10 subplot(2,1,1); // Subplot 1 for the periodic signal
11 plot(t_periodic, x_periodic);
12 xlabel('Time (s)');
13 ylabel('Amplitude');
14 title('Periodic Signal: Sinusoidal Wave (1 Hz)');
15 // Parameters for the aperiodic signal (Exponentially Decaying)
16 a_aperiodic = 0.5; // Base of the exponential decay
17 A_aperiodic = 2; // Amplitude
18 t_aperiodic = 0:0.01:2; // Time vector for 0 to 2 seconds
19 // Generate the aperiodic signal
20 x_aperiodic = A_aperiodic * exp(-a_aperiodic * t_aperiodic);
21 // Plot the aperiodic signal
22 subplot(2,1,2); // Subplot 2 for the aperiodic signal
23 plot(t_aperiodic, x_aperiodic);
24 xlabel('Time (s)');
25 ylabel('Amplitude');
26 title('Aperiodic Signal: Exponentially Decaying Signal');
27
```

